Bikkad IT
Learn Fast, Earn Smart, It's Now !!

WINDOW VALUE FUNCTIONS

An Window Functions

Aggregate
- SUM ()
- AVG ()
- COUNT ()
- MAX ()
- MIN ()

Perform calculations on a set of rows and return a single aggregated value for each row.

Rank
- ROW_NUMBER ()
- RANK ()
- DENSE_RANK ()
- NTILE ()
- CUME_DIST ()
- PERCENT_RANK ()

Assign a rank to each row in a window

Value
- LAG ()
- LEAD ()
- FIRST_VALUE ()
- LAST_VALUE ()

Return a specific value in a window to be compared with the value of current row

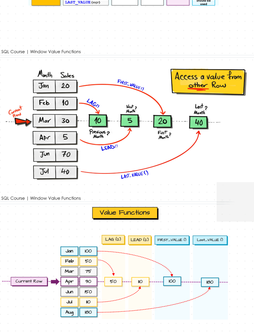
Value Functions

Value Window Functions	LEAD (expr,offset,default)	Returns the value from a previous row
	LAG (expr,offset,default)	Returns the value from a subsequent row
	FIRST_VALUE (expr)	Returns the first value in a window
	LAST_VALUE (expr)	Returns the last value in a window

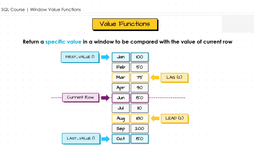
```
LEAD(Sales,2,0) OVER (ORDER BY OrderDate)
LAG (expr,offset,default) OVER (PARTITION BY PartitionExpr ORDER BY OrderExpr)
FIRST_VALUE(Sales) OVER (ORDER BY OrderDate)
FIRST_VALUE(Sales) OVER (ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)
```


Value Functions

	Expression	Partition By	Order By	Frame Clause	
Value Window Functions	LEAD (expr,offset,default)	All Data Type	Optional	Required	Not allowed
	LAG(expr,offset,default)				
	FIRST_VALUE (expr)				Optional
	LAST_VALUE (expr)				Should be used

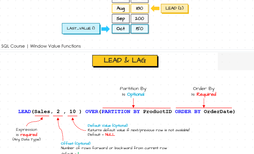


Value Functions



Value Functions

Return a specific value in a window to be compared with the value of current row



LEAD & LAG



LEAD & LAG



LEAD

Access Next Row

```
LEAD(Sales) OVER( ORDER BY Month)
```



LAG

Access Previous Row

```
LAG(Sales) OVER( ORDER BY Month)
```


LEAD

Access Next Row

```
LEAD(Sales) OVER( ORDER BY Month)
```



LAG

Access Previous Row

```
LAG(Sales) OVER( ORDER BY Month)
```


LEAD

Access Next Row

```
LEAD(Sales) OVER( ORDER BY Month)
```



LAG

Access Previous Row

```
LAG(Sales) OVER( ORDER BY Month)
```


LEAD

Access Next Row

```
LEAD(Sales) OVER( ORDER BY Month)
```



LAG

Access Previous Row

```
LAG(Sales) OVER( ORDER BY Month)
```


LEAD & LAG

TIME SERIES ANALYSIS

Year-over-Year (YoY)

Analyze the overall growth or decline of the business's performance over time

Month-over-Month (MoM)

Analyze short-term trends and discover patterns in seasonality

FIRST & LAST

```
FIRST_VALUE(Sales) OVER (ORDER BY Month)
LAST_VALUE(Sales) OVER (ORDER BY Month)
```



Default

```
RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
```


FIRST & LAST

```
LAST_VALUE(Sales) OVER (ORDER BY Month
ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING)
```



SQL WINDOW USE CASES

- Top N Analysis
- Bottom N Analysis
- Identify & Remove Duplicates
- Running Total & Aggregation
- Data Segmentation
- Data Distribution Analysis
- Equality Load Balancing
- Overall Analysis
- Total Per Group Analysis
- Part to Whole Analysis
- Time Series Analysis : MoM & YoY
- Time Gaps Analysis : Customer Retention
- Comparison Analysis : Current vs Target/Previous
- Quality Checks
- Running Total
- Moving Average

OUR LOCATIONS

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Pimpri-Chinchwad
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Nashik

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